W. BLAKE DRECHSLER

Charlottesville, VA
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EDUCATION

PhD, Astronomy *Expected May 2027*
MS, Astronomy *May 2023*
University of Virginia
Advisor: Dr. John Tobin

BA, Astrophysics *May 2020*
University of California Berkeley

TEACHING/GRADING EXPERIENCE

Astronomy 3130 TA Spring 2023
University of Virginia

· Observational Astronomy: Assisted in running the labs, taught a number of lectures, and assisted students through office hours and workshops.

Telescope Observing TA Fall 2021
University of Virginia

· Head TelObs TA: Assisted students during the lab, helped undergraduate TAs operate the telescopes, shuttled the students to and from the observatory, set up and closed down the facility, and graded the labs.

Astronomy 1220/1210 TA Fall 2021
University of Virginia

· Introduction to the Sky and the Solar System/Introduction to the Stars, Galaxies, and the Universe: assisted students via email and office hours, lead review sessions before exams, and graded assignments/exams.

RESEARCH EXPERIENCE

University of Virginia, PI: Dr. John Tobin August 2022-Present

I am using multi-wavelength data from ALMA and JWST to characterize accretion in young protostars and model their masses and disk properties. I also work on developing radiation transfer modeling techniques.

University of Virginia, PI: Professor Jonathan Tan July 2021-August 2022
Graduate Student Researcher

· I studied massive star formation and star cluster formation through numerical simulations. I simulated giant molecular cloud collisions as potential triggers for these star formation events.

UC Berkeley, PI: Professor Jessica Lu Spring 2019-May 2021
Undergraduate/ Post-Baccalaureate Astronomy Researcher

· I ran and analyzed adaptive optics (AO) simulations to test and optimize the performance of the future AO system on the Keck telescope, KAPA. I also I studied the Galactic Center environment by applying Bayesian inference techniques to characterize the orbits of stars around Sgr A*.

PUBLICATIONS

Drechsler W.Blake, (+ 14 co-authors), 2026, "Accretion onto the Embedded Protostar L1527 IRS: Insights from JWST NIRSpec and MIRI" Observations", ApJ, 1000, 299

Devaraj, R. (+13 co-authors and **Drechsler, W.B**), 2026, "JOYS: JWST MIRI/MRS spectra of the inner 500 au region of the L1527 IRS bipolar outflow", arXiv:2601.17820

Slavicinska, K. (+16 co-authors and **Drechsler, W.Blake**, 2025, "HDO Ice Detected toward an Isolated Low-mass Protostar with JWST", ApJL, 986, 2

Finn, M., Mazzei, R., **Drechsler, B.**, Telkamp, Z., Rao, M., Agrawal, P., McAlister, 2023, "How Informal Science Education Influences Elementary Students' Perceptions Of Science And Themselves." Journal of STEM Outreach. 6.10.15695/jstem/v6i1.10.

Jia, S., (14 co-authors + **Drechsler, W.B**, 2023, "Stellar Populations in the Central 0.5 pc of Our Galaxy. III. The Dynamical Substructures", ApJ, 949, 1

INVITED TALKS

Virginia Tech Astronomy Seminar September 2025
"Studying How Stars Grow in the Earliest Stages of Protostellar evolution"

Lorenz Workshop: Protostellar accretion and ejection with JWST and beyond May 2025
"Accretion in L1527 IRS"

POSTER PRESENTATIONS

Discs on the Exe July 2026
"Robust Stellar Mass Estimates for the Edge-On Class 0 Protostar L1527 IRS"

Protostars and Planets VII April 2023
"Determining the Mass and Accretion Mechanism for the Class 0 Protostar L1527 IRS"

From Stars to Galaxies II Summer 2022
"Massive Star Clusters from Colliding Giant Molecular Clouds"

Our Galactic Ecosystem Spring 2022
"Massive Star Clusters from Colliding Giant Molecular Clouds"

Keck Science Meeting Summer 2020
"Performance Simulations of the Keck All-Sky Precision Adaptive Optics System (KAPA)"

AWARDS/ FELLOWSHIPS

Reber Fellowship NRAO

Program giving Ph.D. students an opportunity to conduct research in radio astronomy, radio instrumentation, or computational techniques at one of the NRAO sites under the supervision of an NRAO staff astronomer or engineer

TECHNICAL SKILLS

Python, C/C++, CASA, Git, Unix, Latex